

Rocksdb_dataset-1

Commit 1:

2bf11e1e969db0325459f62623ebb89534db2fe0

Non-C file modifications, therefore no performance regression testing is required.

Commit 2: 2ff9f95f59ae4c941f7148a1ab549e33488a6d47

Non-C file modifications, therefore no performance regression testing is required.

Commit 3: 389f75c86c084a477f5fb3730dd5bb39036dabac

Unable to provide test cases

Commit 4:

a45b9478eee6824928c268dba67d756273b0a34f

Non-C file modifications, therefore no performance regression testing is required.

Commit 5: 09c989fbc4d2c56ca3bcd847b2433fdc2dcaa24

Non-C file modifications, therefore no performance regression testing is required.

Commit 6:

6ae3412244e487a63e9e0c09938230aca611d7a1

Non-C file modifications, therefore no performance regression testing is required.

Commit 7: d957e1a33af7201b6ac3601baa4fd65b0958a25f

Non-C file modifications, therefore no performance regression testing is required.

Commit 8:

1d919ac414141683bac75775b76ea06dab71bfed

Non-C file modifications, therefore no performance regression testing is required.

Commit 9: c72e79a262bf696faf5f8becabf92374fc14b464

The 1st symbolic execution

Testcase:for(int i = 0; i< 1000000; i++){db->SetDBOptions({{"max_background_flushes", "2"}});}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1834875.4553695654
4  The new version is executed by default
5  1836542.8709183673
6  The new version
7  1816339.094478261
8  Comparison of New and old versions (along symbolic execution path)
9  0.010205341583879176
10 Comparison of New and old versions (along the default execution path)
11 0.011123350535991089
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1982536.8598936168
4  The new version is executed by default
5  1981958.5230204083
6  The new version
7  1992848.1135625
8  Comparison of New and old versions (along symbolic execution path)
9  -0.005174129226763054
10 Comparison of New and old versions (along the default execution path)
11 -0.0054643354242512035
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1966151.8349183672
4  The new version is executed by default
5  1960109.4755
6  The new version
7  1956483.1508163265
8  Comparison of New and old versions (along symbolic execution path)
9  0.004941869342450783
10 Comparison of New and old versions (along the default execution path)
11 0.001853491394577251
```

No performance regression

The 4th symbolic execution

Testcase:for (int i = 0; i < 30000; i++){for (int j = 0; j < 100; j++){key = "key_exit" + std::to_string(i + j * 300);keys.emplace_back(key);}values.resize(keys.size());statuses.resize(keys.size());db->MultiGet(rocksdb::ReadOptions(), db->DefaultColumnFamily(), keys.size(), keys.data(), values.data(), statuses.data());keys.clear();}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2810558.238795918
4  The new version is executed by default
5  2838948.204714286
6  The new version
```

```
7 2837849.830265306
8 Comparison of New and old versions (along symbolic execution path)
9 -0.009616996353480954
10 Comparison of New and old versions (along the default execution path)
11 0.0003870445987896123
```

No performance regression

The 5th symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({"write_buffer_size", "131072"});}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 881758.8929148936
4 The new version is executed by default
5 876643.857680851
6 The new version
7 886658.0810425531
8 Comparison of New and old versions (along symbolic execution path)
9 -0.005525453647136352
10 Comparison of New and old versions (along the default execution path)
11 -0.011294346237646758
```

No performance regression

The 6th symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({"write_buffer_size", "131072"});}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 891323.5889787233
4 The new version is executed by default
5 883701.4776249999
6 The new version
```

```
7 886846.1622978724
8 Comparison of New and old versions (along symbolic execution path)
9 0.005048707285657783
10 Comparison of New and old versions (along the default execution path)
11 -0.003545919017932434
```

No performance regression

The 7th symbolic execution

Testcase:for (int i = 0; i < 30000; i++){for (int j = 0; j < 100; j++){key = "key_exit" + std::to_string(i + j * 300);keys.emplace_back(key);}values.resize(keys.size());statuses.resize(keys.size());db->MultiGet(rocksdb::ReadOptions(), db->DefaultColumnFamily(), keys.size(), keys.data(), values.data(), statuses.data());keys.clear();}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2871209.690379999
4 The new version is executed by default
5 2837080.43320000003
6 The new version
7 2838408.0236875
8 Comparison of New and old versions (along symbolic execution path)
9 0.011556360614385874
10 Comparison of New and old versions (along the default execution path)
11 -0.00046772362409509767
```

No performance regression

The 8th symbolic execution

Testcase:for (int i = 0; i < 5000; i++){key = "key_exit" + std::to_string(i + 30001);value = "new_value" + key;db->Put(write_options_sync, key, value);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

```
2 The new version executes along symbols
3 349374.10010416666
4 The new version is executed by default
5 364629.3070425532
6 The new version
7 348215.45077083335
8 Comparison of New and old versions (along symbolic execution path)
9 0.003327392080875334
10 Comparison of New and old versions (along the default execution path)
11 0.04713707055613134
```

No performance regression

The 9th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 1983568.1916808514
4 The new version is executed by default
5 1970543.6417291667
6 The new version
7 1963696.5775208334
8 Comparison of New and old versions (along symbolic execution path)
9 0.010119493198437995
10 Comparison of New and old versions (along the default execution path)
11 0.0034868239252001947
```

No performance regression

The 10th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块
end-to-end diff:

```
2 The new version executes along symbols
3 1984783.1656734694
4 The new version is executed by default
5 1961126.6955000001
6 The new version
7 1963139.7171250002
8 Comparison of New and old versions (along symbolic execution path)
9 0.011024915017340618
10 Comparison of New and old versions (along the default execution path)
11 -0.001025409250009013
```

No performance regression

The 11th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 1701355.5061276592
4 The new version is executed by default
5 1781700.3327755097
6 The new version
7 1691257.8574893621
8 Comparison of New and old versions (along symbolic execution path)
9 0.005970496215927037
10 Comparison of New and old versions (along the default execution path)
11 0.053476455341001386
```

No performance regression

The 12th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  17002259.174333334
4  The new version is executed by default
5  17209406.289473683
6  The new version
7  17033980.943421055
8  Comparison of New and old versions (along symbolic execution path)
9  -0.001862263976523491
10 Comparison of New and old versions (along the default execution path)
11 0.010298552442632714
```

No performance regression

The 13th symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1986102.694857143
4  The new version is executed by default
5  1987980.3136122448
6  The new version
7  2003425.9098333332
8  Comparison of New and old versions (along symbolic execution path)
9  -0.008646795916516452
```

```
10 Comparison of New and old versions (along the default execution path)
11 -0.007709591927147116
```

No performance regression

The 14th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 16826895.18161111
4 The new version is executed by default
5 17018862.485166665
6 The new version
7 17004126.610315792
8 Comparison of New and old versions (along symbolic execution path)
9 -0.01042284809836471
10 Comparison of New and old versions (along the default execution path)
11 0.0008666058062595922
```

No performance regression

The 15th symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
```

```
3 2007375.901680851
4 The new version is executed by default
5 1998175.6059148933
6 The new version
7 1995133.7534782614
8 Comparison of New and old versions (along symbolic execution path)
9 0.006136003754759226
10 Comparison of New and old versions (along the default execution path)
11 0.001524635845255405
```

No performance regression

The 16th symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({{"write_buffer_size", "131072"}});}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 655760.0529583334
4 The new version is executed by default
5 837097.5393829788
6 The new version
7 849871.5574285715
8 Comparison of New and old versions (along symbolic execution path)
9 -0.22840098927131403
10 Comparison of New and old versions (along the default execution path)
11 -0.01503052777085826
```

No performance regression

Commit 10:

2af12ea69c29ecd278b4f46272ea0d29c9c497e5

Non-C file modifications, therefore no performance regression testing is required.

Commit 11:

1acd315c2095c072cc5e52d6b6732c9b2c843d5d

The 1st symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1725133.9185
4  The new version is executed by default
5  1706814.467291667
6  The new version
7  1686703.3050625
8  Comparison of New and old versions (along symbolic execution path)
9  0.02278445374604624
10 Comparison of New and old versions (along the default execution path)
11 0.011923354966344613
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1985834.7569375
4  The new version is executed by default
5  2037616.6544897961
6  The new version
7  2034087.8865416667
8  Comparison of New and old versions (along symbolic execution path)
```

```
9    -0.023722244217385332
10   Comparison of New and old versions (along the default execution path)
11   0.0017348158707778446
```

No performance regression

The 3rd symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1    end-to-end diff:
2    The new version executes along symbols
3    17251586.976611115
4    The new version is executed by default
5    17242843.813444447
6    The new version
7    17665033.4893
8    Comparison of New and old versions (along symbolic execution path)
9    -0.023404796426755584
10   Comparison of New and old versions (along the default execution path)
11   -0.023899738209456654
```

No performance regression

The 4th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1722374.2360851066
4  The new version is executed by default
5  1694539.9645625
6  The new version
7  1686656.860020408
8  Comparison of New and old versions (along symbolic execution path)
9  0.021176433044161995
10 Comparison of New and old versions (along the default execution path)
11 0.004673804571011896
```

No performance regression

The 5th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1694972.590265306
4  The new version is executed by default
5  1708233.8779787228
6  The new version
7  1701089.0352708334
8  Comparison of New and old versions (along symbolic execution path)
9  -0.003595605449631068
10 Comparison of New and old versions (along the default execution path)
11 0.004200157992760117
```

No performance regression

Commit 12:

c8bc2b63fce15583269d45029ce49c30ed9f4325

Unable to provide test cases

Commit 13:

f7b42166283223d29b6876ecb6acfb6373f07eaf

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  255144.53231249997
4  The new version is executed by default
5  253244.90820930226
6  The new version
7  265898.41583333333
8  Comparison of New and old versions (along symbolic execution path)
9  -0.04044357875217263
10 Comparison of New and old versions (along the default execution path)
11 -0.04758775107544216
```

No performance regression

The 2nd symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  126388.19608163265
4  The new version is executed by default
5  273838.4263877551
6  The new version
7  276829.83687755105
8  Comparison of New and old versions (along symbolic execution path)
9  -0.5434444584904431
10 Comparison of New and old versions (along the default execution path)
11 -0.01080595402409293
```

No performance regression

The 3rd symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  270171.08792
4  The new version is executed by default
5  271066.2600612245
6  The new version
7  272528.5828541667
8  Comparison of New and old versions (along symbolic execution path)
9  -0.008650450200404076
10 Comparison of New and old versions (along the default execution path)
11 -0.005365759354954253
```

No performance regression

The 4th symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  266582.796
4  The new version is executed by default
5  274830.4818913043
6  The new version
7  273440.27740425535
8  Comparison of New and old versions (along symbolic execution path)
9  -0.02507853440375664
10 Comparison of New and old versions (along the default execution path)
11 0.005084124768472488
```

No performance regression

Commit 14: cc30226b1fa63bc9e65a15c3dbef8e34784ee70c

The 1st symbolic execution

```
Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2039606.0219999999
4  The new version is executed by default
5  2038541.7300204085
6  The new version
7  2022527.3779375001
```

```
8   Comparison of New and old versions (along symbolic execution path)
9   0.008444209086512307
10  Comparison of New and old versions (along the default execution path)
11  0.00791799026188671
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1   end-to-end diff:
2   The new version executes along symbols
3   1993884.1671041667
4   The new version is executed by default
5   1984988.5245918366
6   The new version
7   1972038.696212766
8   Comparison of New and old versions (along symbolic execution path)
9   0.011077607621673043
10  Comparison of New and old versions (along the default execution path)
11  0.0065667212331788054
```

No performance regression

Commit 15: cf768a2f9e5451a99aa774d1ef2eb19681b8df6f

Unable to provide test cases

Commit 16:

54b614de5bd3e26d332b85557d44bde86b2a2e87

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1987364.5483958332
4  The new version is executed by default
5  1990995.9854468089
6  The new version
7  1992272.068369565
8  Comparison of New and old versions (along symbolic execution path)
9  -0.002463278008885627
10 Comparison of New and old versions (along the default execution path)
11 -0.0006405163948318196
```

No performance regression

Commit 17:

19e4aba3db75bd6add7177164c892ab6cdfd50b3

Non-C file modifications, therefore no performance regression testing is required.

Commit 18:

18cecb9c46b4c2a8b148659dac2fcab5a843d32b

The 1st symbolic execution

```
Testcase:for (int i = 0; i < 100; i++){rocksdb::BackupEngine
*backup_engine;rocksdb::BackupEngine::Open(rocksdb::Env::Default(),
rocksdb::BackupEngineOptions("/tmp/test"), &backup_engine);backup_engine-
>CreateNewBackup(db);delete backup_engine;}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  270740.77722916665
4  The new version is executed by default
5  277786.1468367347
6  The new version
```

```
7 274940.3920612245
8 Comparison of New and old versions (along symbolic execution path)
9 -0.015274637533515514
10 Comparison of New and old versions (along the default execution path)
11 0.010350442705692102
```

No performance regression

Commit 19:

30d516229867ea44b42515a07ef579e79e8070d0

The 1st symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2043747.5091632656
4 The new version is executed by default
5 1980091.1844489796
6 The new version
7 2000781.8250208332
8 Comparison of New and old versions (along symbolic execution path)
9 0.021474447441057212
10 Comparison of New and old versions (along the default execution path)
11 -0.010341277751080217
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i < 100; i++){rocksdb::Slice start_key("key_exit:300"), end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db->DefaultColumnFamily(), start_key, end_key);rocksdb::Slice start_key1("key_exit:200");rocksdb::Slice end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db->CompactRange(comp_options, &start_key1, &end_key2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1677013.2347446806
4  The new version is executed by default
5  1714236.2254583333
6  The new version
7  1708628.2063409092
8  Comparison of New and old versions (along symbolic execution path)
9  -0.018503131037461505
10 Comparison of New and old versions (along the default execution path)
11 0.00328217636617027
```

No performance regression

The 3rd symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  16975146.460368425
4  The new version is executed by default
5  17113359.116315793
6  The new version
7  16934738.065894734
8  Comparison of New and old versions (along symbolic execution path)
9  0.002386124563394928
10 Comparison of New and old versions (along the default execution path)
11 0.010547612235041784
```

No performance regression

The 4th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1704620.8001666667
4  The new version is executed by default
5  1709640.532106383
6  The new version
7  1688513.2305208333
8  Comparison of New and old versions (along symbolic execution path)
9  0.009539498627952692
10 Comparison of New and old versions (along the default execution path)
11 0.012512369582696645
```

No performance regression

Commit 20:

e3024e7b58fad26da5dc271a87121908cc79a8ca

Non-C file modifications, therefore no performance regression testing is required.